

HOW MUCH DOES BODY POSE HELP PEDESTRIAN TRAJECTORY FORECASTING? A SEED-REPLICATED STUDY AND AUXILIARY-LOSS AUDIT ON WAYMO

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ABSTRACT

Body pose is a plausible but rarely used signal for pedestrian trajectory forecasting in autonomous driving. We extend the Motion Transformer (MTR) with a pose-conditioning branch that encodes past SMPL body poses (6D rotation representation) via cross-attention with sinusoidal positional encoding, and evaluate it on a pedestrian subset of the Waymo Open Motion Dataset using LiDAR-derived pose pseudo-ground-truth. Conditioned on HD map context, this pose encoder reduces minADE by 1.6% relative to a matched trajectory-only model, an effect that holds across five random seeds ($p < 10^{-4}$) and across two control ablations designed to rule out a heading confound: zeroing the heading-coupled root-orientation channel and switching to an agent-centric pose frame both leave the benefit intact. The improvement is therefore genuine body-configuration signal, not re-encoded heading. We establish this result using paired bootstrap confidence intervals over 6,688 validation pedestrians and three to five seeds per configuration, a replication protocol that we show is necessary for pose-conditioned ablations of this kind. Without it, several plausible-looking effects would be mistaken for real signal when they are in fact seed noise, including a positional-encoding ordering effect and, in one case, a widely used geodesic $SO(3)$ supervision loss that we show receives identically zero gradient under sparse future-pose annotation. We characterize the benefit’s boundaries through the same seed-replicated protocol: pose conditioning does not yet help without map context or after full-Waymo backbone pretraining, and auxiliary-loss composition is not what drives the benefit we find. Together, these results give a calibrated estimate of pose’s value for pedestrian trajectory forecasting, together with a replication and auxiliary-loss-auditing protocol for evaluating pose-conditioned models with confidence.

1 INTRODUCTION

Predicting where pedestrians will walk is a safety-critical problem in autonomous driving. State-of-the-art trajectory forecasting models such as MTR (Shi et al., 2022; 2024) achieve impressive accuracy by operating on observed agent positions, velocities, and HD map geometry, but they discard a potentially rich source of information: the *body pose* of the pedestrian.

Human gait encodes intent. A pedestrian who is turning to face a crosswalk will likely cross it; one whose torso is oriented away from traffic is unlikely to step into the road. A runner’s body orientation predicts turns before the feet complete them. Modern perception systems increasingly provide body pose estimates (Zhu et al., 2023). While recent work uses skeleton keypoints for pose-conditioned prediction in robot navigation and pedestrian benchmarks (Salzmann et al., 2023; Saadatnejad et al., 2024; Gao et al., 2025), state-of-the-art autonomous driving forecasters operating on Waymo (Ettinger et al., 2021) discard body pose entirely. This paper asks: *does conditioning on past SMPL body poses improve pedestrian trajectory prediction within a SOTA AV forecasting framework, and if so, what design choices drive the gain?*

We extend MTR with a *pose conditioning branch* that encodes the past 11 timesteps of SMPL body pose (Loper et al., 2015) using 6D rotation representations (Zhou et al., 2019), a GRU encoder, and

auxiliary future-pose prediction heads. We train and evaluate on a pedestrian subset of the Waymo Open Motion Dataset (Ettinger et al., 2021) paired with LiDAR-derived SMPL pose pseudo-ground-truth from Waymo-3DSkelMo (Zhu et al., 2025), using minADE and minFDE over 6 predicted modes.

This paper reports where and why body pose helps pedestrian trajectory forecasting, established with a replication protocol strong enough to keep a small effect honest. We train and evaluate every configuration with three to five random seeds and paired bootstrap confidence intervals over 6,688 validation pedestrians (Section 4.1). Our findings:

- **Conditioned on HD map context, pose conditioning gives a real, seed-stable improvement, and it is genuine body-configuration signal.** A cross-attention encoder with sinusoidal positional encoding over past SMPL poses reduces minADE by 1.6% relative to the matched trajectory-only model, stable across five seeds ($p < 10^{-4}$). Two control ablations rule out the obvious confound: the benefit is unchanged when the heading-coupled root-orientation channel is zeroed, and again under an agent-centric pose frame. With the map already supplying routing, the residual pose signal reflects body configuration rather than re-encoded heading.
- **This benefit is specific, and we map its boundaries with the same seed-replicated protocol.** It depends on both map context and the cross-attention encoder. A GRU encoder gains nothing from map context, and the benefit does not survive full-Waymo backbone pretraining at five seeds. Without map context, no pose configuration reliably beats the no-pose baseline, and neither auxiliary-loss composition nor dense future-pose supervision changes that picture. We detail these boundary ablations in Sections 4.5 through A.5.
- **Seed replication is what makes this boundary trustworthy, and it surfaces a general auxiliary-loss pitfall.** At three or fewer seeds, several architectural effects in the no-map regime look stable and then reverse once further seeds are added (Section 4.2); the spread among replicates of a single configuration (0.27 minADE across ten seeds’ extremes) exceeds every architectural effect we measure except map and pretraining. Separately, we show analytically and empirically that a widely used geodesic $SO(3)$ supervision loss receives identically zero gradient whenever future pose targets are absent or degenerate, a condition that follows naturally from sparse pose annotation (Section 4.3). We describe both the replication protocol and the gradient check so they transfer to other pose-conditioned forecasting work.

2 RELATED WORK

2.1 TRAJECTORY FORECASTING FOR AUTONOMOUS DRIVING

Modern trajectory prediction methods model multi-agent interactions over HD map features using transformer architectures (Vaswani et al., 2017; Shi et al., 2022; 2024; Yuan et al., 2021; Salzmann et al., 2020). Social GAN (Gupta et al., 2018) introduced generative multimodal prediction for pedestrians; MTR (Shi et al., 2022) advances this with global intention localization and local refinement via cross-attention over map and agent context. None of these methods incorporate body pose information.

2.2 HUMAN POSE ESTIMATION AND POSE REPRESENTATIONS

SMPL (Loper et al., 2015) provides a parametric body model where each joint rotation is represented as a 3D rotation matrix. For neural network learning, Zhou et al. (2019) demonstrate that 6D rotation representations (the first two columns of $SO(3)$) are continuous and better conditioned than quaternions or axis-angle, particularly under gradient-based optimization. The AMASS dataset (Mahmood et al., 2019) provides a large-scale archive of motion-captured body animations in SMPL parameterization and is widely used to train human motion priors (Rempe et al., 2021; He et al., 2022). Waymo-3DSkelMo (Zhu et al., 2025) combines LiDAR-based human mesh recovery (Fan et al., 2023) with such priors to produce temporally coherent SMPL pose sequences for Waymo pedestrians; we use its output as pose pseudo-ground-truth. Works such as MotionBERT (Zhu et al., 2023) and HumanMAC (Chen et al., 2023) focus on pose quality but do not connect body pose to downstream

trajectory prediction. A separate line uses 2D pose for pedestrian *crossing-intention* classification (Fang & López, 2018; Rasouli et al., 2019), supporting the premise that body configuration carries intent, though without producing metric trajectory forecasts.

2.3 JOINT POSE AND TRAJECTORY MODELS

Several recent works condition trajectory prediction on body pose. Salzmann et al. (2023) leverage 3D skeletal keypoints from RGB-D cameras to improve trajectory prediction in robot navigation scenarios. Saadatnejad et al. (2024) integrate 2D/3D skeleton pose cues into a transformer-based predictor on pedestrian benchmarks, and Gao et al. (2025) extend this line to social navigation datasets using skeleton-based body language features. In contrast, our work uses parametric SMPL body pose in 6D rotation space, a richer, anatomically-grounded representation, within the state-of-the-art MTR framework on Waymo, and isolates which design choices actually drive the gain using seed-replicated experiments and per-pedestrian confidence intervals: auxiliary pose-loss design, temporal encoding (GRU vs. cross-attention $\pm$ positional encoding), future-pose supervision quality, and the interaction of pose with HD map context and backbone pretraining.

3 METHOD

3.1 BACKGROUND: MOTION TRANSFORMER (MTR)

MTR (Shi et al., 2022) is a transformer-based trajectory predictor for the Waymo Open Motion Dataset. For each *center object* (the agent being predicted), it encodes a scene context tensor from agent histories and HD map polylines via a PointNet-like encoder, then decodes $K = 6$ trajectory modes through iterative cross-attention over the scene context, guided by K learned intention queries that are globally localized to trajectory clusters.

The loss has three components: a mode classification loss $\mathcal{L}_{\text{cls}}$, a winner-takes-all regression loss $\mathcal{L}_{\text{reg}}$, and a velocity consistency loss $\mathcal{L}_{\text{vel}}$.

3.2 POSE CONDITIONING BRANCH

We add a pose conditioning branch alongside the trajectory decoder, illustrated in Figure 1. The branch has four components:

6D Rotation Encoding. Each SMPL body pose at timestep t consists of 24 joint rotations. We represent each rotation as a 6D vector (Zhou et al., 2019) (the first two columns of the 3×3 rotation matrix), giving a 144D pose vector per timestep.

GRU Pose Encoder. Given the past $T = 11$ pose vectors $\mathbf{p}_1, \dots, \mathbf{p}_T \in \mathbb{R}^{144}$, a two-layer GRU with hidden size 256 processes the sequence:

$$\mathbf{h}_t = \text{GRU}(\mathbf{p}_t, \mathbf{h}_{t-1}), \quad \mathbf{h}_T \in \mathbb{R}^{256}. \quad (1)$$

The final hidden state $\mathbf{h}_T$ is concatenated with the MTR agent feature and projected to 256D via an MLP, forming the fused agent representation.

Pose Prediction Heads. Each of the decoder layers additionally predicts K modes of future poses: at layer ℓ and mode k , a linear head outputs 6D rotations for all 80 future timesteps (a 80×144 output). These heads are auxiliary: their predictions never feed back into the trajectory pathway, and their only effect on trajectory prediction is through the gradients their losses send into the shared decoder features and the GRU encoder.

Auxiliary Pose Losses. Let $\hat{\mathbf{r}}^{ij} \in \mathbb{R}^6$ be the predicted 6D rotation and $\mathbf{r}^{ij} \in \mathbb{R}^6$ be the ground-truth 6D rotation for joint j at future step i . We consider four auxiliary signals:

- **Winner-takes-all L1 (WTA-L1):** $\mathcal{L}_{\text{wta}}$ selects, per pedestrian, the pose mode closest to the ground truth in 6D L2 distance, and applies an L1 regression loss to that mode, masked by per-timestep trajectory validity. (In our configuration files this loss is named `gmm_pose` for historical reasons; it is a plain WTA regression, not a Gaussian mixture likelihood.)

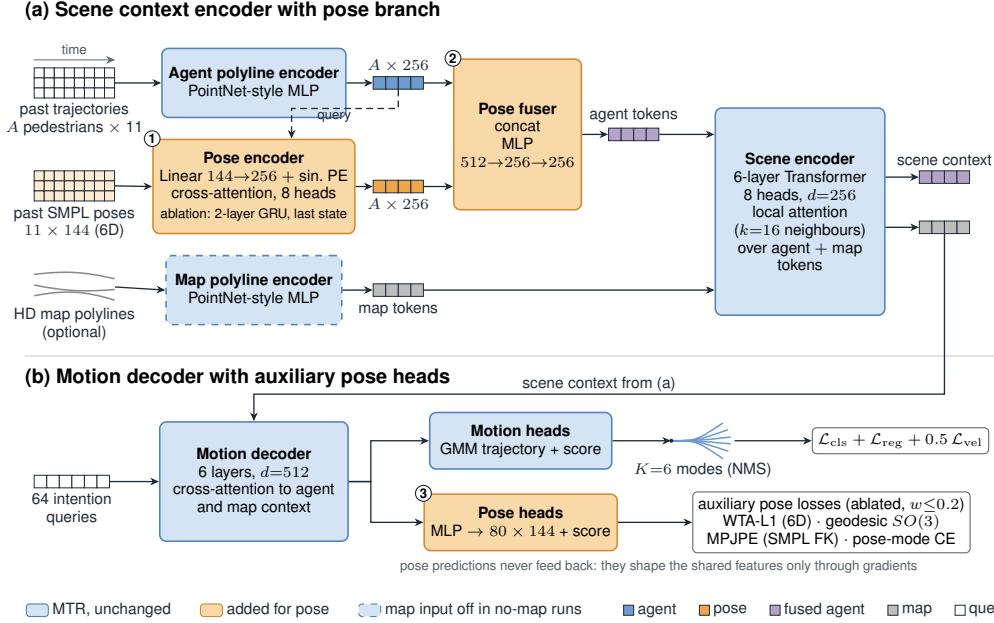


Figure 1: **MTR+Pose architecture.** (a) The pose encoder (1) embeds each pedestrian’s past 11 SMPL poses (24 joints in 6D rotation, 144D per frame) and, in our main configuration, cross-attends to them with the agent’s PointNet feature as query, after adding a sinusoidal positional encoding; the GRU variant uses the last hidden state instead. The pose fuser (2) concatenates the pose and agent features and projects them back to 256D *before* MTR’s scene encoder, which then attends jointly over agent and (optional) HD map tokens. (b) After every decoder layer, pose heads (3) predict 80 future poses and a pose score for each intention query, trained with the auxiliary losses of Section 3.2. Their predictions never feed back into the trajectory pathway, so pose supervision shapes the shared features only through gradients. All other MTR modules are unchanged.

- **Geodesic loss:** $\mathcal{L}_{geo} = \frac{1}{IJ} \sum_{i,j} \arccos \left(\text{clamp} \left(\frac{\text{tr}(\mathbf{R}_i^j \mathbf{R}_i^j) - 1}{2}, -1 + \epsilon, 1 - \epsilon \right) \right)$, where $\mathbf{R}, \hat{\mathbf{R}} \in SO(3)$ are rotation matrices recovered from the 6D representations of the WTA-selected mode, averaged over all future steps (unmasked). The clamp ($\epsilon=10^{-7}$) avoids the divergent arccos derivative at the ± 1 boundary.
- **MPJPE:** $\mathcal{L}_{mpjpe} = \frac{1}{IJ} \sum_{i,j} \|\mathbf{x}_i^j - \hat{\mathbf{x}}_i^j\|_1$, where $\mathbf{x}$ are joint positions computed via forward kinematics on SMPL.
- **Pose mode classification:** $\mathcal{L}_{cls,pose}$, a cross-entropy loss on per-mode pose scores against the WTA-selected mode.

The total loss adds the three MTR motion terms ($\mathcal{L}_{cls} + \mathcal{L}_{reg} + 0.5 \mathcal{L}_{vel}$) to the weighted pose terms $w_{wta} \mathcal{L}_{wta} + w_{geo} \mathcal{L}_{geo} + w_{mpjpe} \mathcal{L}_{mpjpe} + w_{cls,pose} \mathcal{L}_{cls,pose}$; we ablate $w_* \in \{0, 0.05, 0.1, 0.2\}$ (weights of 1.0 diverge to NaN through the SMPL forward pass, so all stable runs use $w \leq 0.2$).

The geodesic loss term is inert in the main data condition. In our primary (10 fps) data condition, SMPL pose observations cover only the *past* window ($t \leq 1.0$ s); future pose targets are zero for 99.99% of timesteps (5 of 91,520 in the training split). As we show analytically and empirically in Section 4.3, this renders the geodesic term *inert* (identically zero gradient against all-zero targets, which map to the all-zero matrix under Gram-Schmidt), while WTA-L1, MPJPE, and $\mathcal{L}_{cls,pose}$ stay active but pull predictions toward the degenerate zero targets. We flag this prominently because it is easy to miss: the loss degrades silently to a constant rather than erroring, and left unchecked it invites mistaking single-run seed noise for a supervision effect (Section 4.3).

3.3 CROSS-ATTENTION POSE ENCODERS (ABLATION)

To isolate the contribution of temporal ordering from sequential integration, we test two cross-attention pose encoder variants. Let $\mathbf{q} \in \mathbb{R}^{256}$ be the MTR agent feature (projected query) and $\mathbf{K}, \mathbf{V} \in \mathbb{R}^{11 \times 256}$ be the 11 projected pose frames (keys and values). The cross-attention output is:

$$\mathbf{c} = \text{softmax}\left(\frac{\mathbf{q}\mathbf{K}^\top}{\sqrt{256}}\right)\mathbf{V}, \quad (2)$$

and $\mathbf{c}$ is concatenated with $\mathbf{q}$ and projected to 256D, replacing $\mathbf{h}_T$ in the fusion step.

Cross-attention without positional encoding. No temporal information is injected before attention: $\mathbf{K} = \{W_K \mathbf{p}_t\}_{t=1}^{11}$, $\mathbf{V} = \{W_V \mathbf{p}_t\}_{t=1}^{11}$. This treats pose frames as a permutation-invariant set and tests whether the raw content of each pose frame, absent any ordering, is sufficient to condition trajectory prediction.

Cross-attention with sinusoidal positional encoding. Standard sinusoidal positional encoding (Vaswani et al., 2017) is added to the projected keys and values before attention: $\mathbf{K}' = \{W_K \mathbf{p}_t + \mathbf{e}_t\}_{t=1}^{11}$, where $\mathbf{e}_t$ is the fixed sinusoidal embedding for frame index $t \in \{0, \dots, 10\}$. This injects temporal ordering at zero extra parameter cost and tests whether ordering alone is sufficient, without the causal accumulation structure of a GRU.

4 EXPERIMENTS

4.1 DATASET AND SETUP

Dataset. We use a pedestrian-only subset of the Waymo Open Motion Dataset (Ettinger et al., 2021) paired with SMPL body pose pseudo-ground-truth from the Waymo-3DSkelMo pipeline (Zhu et al., 2025). The subset contains 579 training scenes (1,348 pedestrians, of which 1,144 have sufficient valid past observations to serve as prediction targets) and 5,171 validation scenes (8,282 pedestrians). Ground-truth future trajectories come from Waymo tracking data (91 timesteps at 10 Hz; 9.0 s total span, comprising 1.0 s past plus an 8.0 s future horizon). 6,688 of the 8,282 validation pedestrians (80.8%) have valid future trajectory data and enter the evaluation.

Pose provenance. The SMPL poses are not motion-capture ground truth: they are estimated from Waymo LiDAR by the Waymo-3DSkelMo pipeline (Zhu et al., 2025), which fits per-frame SMPL bodies with LiDAR-HMR (Fan et al., 2023) and refines them with AMASS-trained motion priors (Mahmood et al., 2019; Rempe et al., 2021; He et al., 2022). Two properties matter for our results. *First*, pose observations exist only for the past window in our main condition, so no future information leaks into the inputs; the evaluation is sound. *Second*, the pipeline snaps the SMPL root rotation to the past-trajectory motion direction whenever the pedestrian is moving and the raw LiDAR estimate deviates by more than 90° , so the root-orientation channel is partially derived from position history the model already receives, an overlap we quantify in Section 4.4. Body configuration and gait within $\pm 90^\circ$ are genuinely perception-derived.

Evaluation. We report **minADE** (minimum Average Displacement Error over 6 modes, in meters) and **minFDE** (minimum Final Displacement Error). Lower is better. The evaluation covers the full 8.0 s future horizon (80 steps at 10 Hz). As an externally interpretable anchor we also compute **MR@2m** (fraction of pedestrians with best-checkpoint minFDE > 2 m): 0.239 no-map, 0.145 map, 0.093 pretrained; pose moves it by ≤ 0.005 in any cell. Official Waymo metrics (mAP, gated miss rate) require full retained predictions, which our per-epoch protocol does not store.

Training and statistical protocol. All models train for 30 epochs with batch size 10 on a single RTX 4090 (AdamW, LR 10^{-4} , step decay $\times 0.5$ at epochs 10/20/25, gradient-norm clip 1.0); a full run takes 10–15 minutes. Validation is run at epochs 1, 2, 4, every second epoch up to 20, and every epoch from 21 to 30. Following common practice, our per-run headline number is the *best* validation minADE over these evaluations; because best-of- n selection on a noisy curve inflates apparent differences, every key comparison is additionally run with **three to five random seeds** (five for the headline, encoder, map, and pretrain cells; three for secondary cells), and we report mean $\pm$ std across seeds together with **paired bootstrap** 95% confidence intervals over the 6,688

Table 1: Main result (no map; best-checkpoint protocol; 5 seeds each). Per-seed values expose a heavy-tailed failure mode in *both* GRU- and attention-based pose conditioning (bold = degraded seeds) that mean $\pm$ std alone understates. No row is significantly below the baseline under a seed-resampling bootstrap.

Model	Aux. loss	minADE $\downarrow$ (mean $\pm$ std)	per-seed values
MTR (no pose)	—	0.6604 ± 0.0143	.655 .683 .664 .654 .646
MTR+POSE GRU, no aux	none active	0.6545 ± 0.0271	.696 .665 .639 .626 .647
MTR+POSE GRU, WTA-L1	$w=0.1$	0.6853 ± 0.0651	.639 .785 .629 .717 .657
MTR+POSE XAttn+PE, WTA-L1	$w=0.1$	0.6648 ± 0.0381	.640 .634 .637 .710 .703

evaluated pedestrians (per-pedestrian errors averaged across seeds, resampled jointly for both models in a pair; 10,000 resamples).

Scale caveat. The 579-scene training subset is a small fraction of the full Waymo training set ($\approx 487k$ scenarios), so absolute minADE values in Sections 4.2–4.5 are far from full-data state of the art; relative, seed-replicated comparisons between models trained on identical data are the object of study. Section 4.6 quantifies the scale effect directly.

4.2 MAIN RESULT: POSE CONDITIONING VS. BASELINE

Table 1 compares, over five seeds each, the trajectory-only MTR baseline against three pose-conditioned variants: the GRU encoder with the WTA-L1 auxiliary pose loss ($w=0.1$; see Section 4.3 for why the nominal “geodesic” supervision in this configuration is inert), the GRU encoder with *no* active auxiliary pose loss, and the cross-attention encoder with sinusoidal positional encoding (same WTA-L1 auxiliary loss).

At five seeds, neither pose-conditioned variant reliably beats the baseline, and per-seed values expose why single-run comparisons are misleading here. The GRU encoder with the WTA-L1 auxiliary loss is *worse than baseline on average*: two of five seeds converge to degraded optima (bold), and the pedestrian bootstrap puts it at +3.8% minADE, even though its single best seed (0.629) would look like a -4.8% win in isolation. The cross-attention encoder with positional encoding looks strong at three seeds, a remarkably stable -3.5% ($\sigma 0.0026$), but two additional seeds, run under identical code, both fail (best-epoch 0.710/0.703). The five-seed cell is 0.6648 ± 0.0381 , a fifteen-fold spread increase, on the *wrong* side of the baseline (+0.7%, $p=0.83$). The attention encoder is heavy-tailed too; its three-seed stability was itself a sampling artifact, exposed only once the fourth and fifth seeds ran. The apparent positional-encoding effect dissolves with it (no-PE vs. PE: $p=0.76$ at five seeds). Under seed resampling, *no* no-map configuration reliably beats the no-pose baseline; pose’s benefit, if any, must be sought where the backbone is already strong (Sections 4.5–4.6).

4.3 THE AUXILIARY-LOSS AUDIT: WHY THE GEODESIC TERM IS INERT

A natural hypothesis, given the losses defined in Section 3.2, is that geodesic $SO(3)$ supervision is the critical ingredient of the pose branch: single-run comparisons put “geodesic-only” supervision at 0.6231 (-7.6% vs. baseline 0.6745), ahead of “WTA-L1 only” at 0.6467. Three facts show this hypothesis is wrong.

Fact 1: the geodesic term received zero gradient. In the 10 fps data condition, future pose targets are zero for all but 5 of 91,520 training steps (Appendix A.2). A zero 6D target maps under Gram–Schmidt to the all-zero matrix $\mathbf{R}_{\text{gt}}=\mathbf{0}$, for which $\text{tr}(\dot{\mathbf{R}}^\top \mathbf{R}_{\text{gt}}) \equiv 0$ for *any* prediction, so $\mathcal{L}_{\text{geo}} \equiv \arccos(-\frac{1}{2})$, a constant; we verified its gradient is exactly zero (and nonzero under random targets).

Fact 2: the two compared configurations were actually identical. The “geodesic-only” run also carried WTA-L1 at $w=0.1$, so with the geodesic term inert it ran the same training distribution as

Table 2: Pose supervision ablation, seed-replicated (no map, GRU encoder, $w=0.1$ for active losses). “No aux” has no active auxiliary pose loss (the inert geodesic term only), so the pose encoder is trained by trajectory gradients alone. The WTA-L1 row pools the single-run “geodesic-only” and “WTA-L1 only” configurations, which are functionally identical, with eight further seeds. Single-run values shown for comparison.

Aux. pose loss	minADE (mean $\pm$ std)	per-seed values	single-run
None (no pose)	0.6604 ± 0.0143	.655 .683 .664 .654 .646	0.6745
No aux (inert geo)	0.6545 ± 0.0271	.696 .665 .639 .626 .647	—
WTA-L1 (10 seeds, pooled)	0.6957 ± 0.0841	.639 .785 .629 .717 .657 .897 .662 .662 .659 .650	$0.6467 / 0.6231^\dagger$
MPJPE + WTA-L1	0.6723 ± 0.0321	.724 .662 .644 .680 .651	0.6576
All (MPJPE+WTA+cls)	0.6618 ± 0.0366	.668 .674 .630 .623 .714	0.6532

† “WTA-L1 only” and “geodesic-only” were run once each and initially treated as two configurations; because the geodesic term is inert (Fact 1), they are the same configuration, and their spread is seed noise.

“WTA-L1 only”. Their 0.0236 gap (0.6231 vs. 0.6467) is pure run-to-run noise, larger than most of the architectural effects we measure elsewhere in this paper.

Fact 3: seed replication measures the noise directly. Re-run with multiple seeds (Table 2), the pooled ten replicates of this single configuration span minADE 0.629–0.897 (0.6957 ± 0.0841); both single-run values above sit comfortably inside this one-configuration spread, confirming that the apparent “geodesic effect” is a draw from ordinary seed noise, not a supervision effect.

Seed replication reverses a second, related expectation: far from being “the critical ingredient”, the auxiliary losses, with the GRU encoder and degenerate zero-valued targets, are associated with *instability*. Four of fifteen runs across the WTA-L1 and MPJPE+WTA-L1 cells converged badly (≥ 0.71), whereas the no-auxiliary control produced no such seed in five runs. At our statistical power the active-loss variants cannot be usefully ranked against one another, and any single-run mechanistic story about geodesic gradients shaping the GRU would be fitted to noise.

The mask fix tames the worst failures, but adds no signal. Because the audit shows the pose losses train against absent targets, we re-ran the audit cells (five seeds each) with a corrected future-pose validity mask (losses averaged only over the $< 0.01\%$ of steps that carry a real target; this is the released code). Masking removes the WTA-specific catastrophic tail: the masked WTA-L1 cell’s worst seed is 0.679 vs. 0.897 unmasked (σ 0.065 $\rightarrow$ 0.018). It does not, however, turn any auxiliary loss into a benefit: at five seeds *no* masked cell differs from the masked baseline under seed resampling (all hierarchical $p \geq 0.15$), degraded seeds (≥ 0.70) still occur even with no active auxiliary, and the fix leaves the no-pose baseline unchanged (0.6605 ± 0.0153 masked vs. 0.6604 ; $p=0.90$). The on-10fps auxiliary pose signal is, as predicted, inert; the no-map regime is noisy with or without it.

Reproducibility note. All seed-matrix results were produced with the *pre-fix* code (tagged in the release); the released code masks the future-pose losses, so re-running any aux-active cell yields the masked training distribution. The baseline is verified unchanged ($p=0.95$) and the four audit cells are re-run masked above; other aux-active cells were not, and may differ under the fix.

4.4 ABLATION: TEMPORAL ENCODING, GRU VS. CROSS-ATTENTION

Table 3 compares three temporal encoding strategies (three to five seeds each), all with the WTA-L1 auxiliary loss at $w=0.1$.

At three seeds this ablation seems to expose a clean mechanism: bag-of-frames useless, +PE best-in-class, GRU worst, suggesting temporal *ordering* carries the pose signal. At five seeds the mechanism dissolves with the main effect: no-PE and +PE are indistinguishable ($p=0.76$), every variant overlaps the baseline, and each has heavy-tailed seeds. In the no-map regime no encoder property is resolvable: any three-seed ranking among GRU, no-PE attention, and +PE attention is a draw from wide, overlapping seed distributions, not a stable architectural effect.

Table 3: Temporal encoding ablation, seed-replicated (no map; WTA-L1 auxiliary loss at $w=0.1$ in all pose-conditioned variants).

Pose Encoder	Temporal info	minADE (mean $\pm$ std)	per-seed values
None (no pose)	—	0.6604 ± 0.0143	.655 .683 .664 .654 .646
Cross-attention, no PE	none (bag)	0.6705 ± 0.0336	.714 .641 .646 .653 .700
Cross-attention + sinus. PE	ordering only	0.6648 ± 0.0381	.640 .634 .637 .710 .703
GRU (sequential)	ordering + causal	0.6853 ± 0.0651	.639 .785 .629 .717 .657

Table 4: Map $\times$ pose ablation plus mechanism controls, seed-replicated (5 seeds per cell; the two control cells 3). WTA-L1 auxiliary loss at $w=0.1$ in pose cells. With map context the seed variance collapses by an order of magnitude relative to the no-map cells.

Pose Enc.	Map	minADE (mean $\pm$ std)	per-seed values
None	No map	0.6604 ± 0.0143	.655 .683 .664 .654 .646
None	With map	0.4854 ± 0.0047	.490 .484 .489 .478 .486
GRU	No map	0.6853 ± 0.0651	.639 .785 .629 .717 .657
GRU	With map	0.4823 ± 0.0058	.480 .491 .485 .477 .479
XAttn+PE	With map	0.4778 ± 0.0041	.477 .481 .480 .471 .480
+ root-orient zeroed	With map	0.4800 ± 0.0046	.477 .485 .478
+ agent-centric pose frame	With map	0.4801 ± 0.0056	.475 .486 .479

Root-orientation control. A control ablation asks *what* in the pose an encoder could use: zeroing the root-orientation channel of the xattn+PE input (joint 0; body joints untouched) gives 0.6628 ± 0.0056 , indistinguishable from both the no-pose baseline ($p=0.72$) and the five-seed xattn+PE cell ($p=0.88$). With no established no-map gain, this ablation is uninformative here; it becomes decisive in the map condition (Section 4.5), where a real pose benefit exists to decompose. The channel matters because the Waymo-3DSkelMo pipeline partially derives it from past-trajectory heading (Section 4.1), so a benefit that survives its removal cannot be heading re-entry.

4.5 ABLATION: HD MAP CONTEXT

The experiments above use zero-padded map features (the SMPL-paired pedestrian subset was originally processed without HD map polylines). To quantify how map context interacts with pose conditioning, we enable HD map loading and train the 2×2 factorial design: $\pm \text{map} \times \pm \text{pose}$ (GRU encoder, WTA-L1 auxiliary loss at $w=0.1$), three seeds per cell.

Table 4 reveals three findings. *First*, HD map is the dominant signal: it alone reduces minADE by 26.5% (CI $[-29.0, -24.1]$), far more than any pose effect, since routing is encoded directly in the map. *Second*, this is where pose’s one seed-significant benefit lives, and it requires the attention encoder. The GRU cell does not separate from no-pose+map at five seeds (-0.6% , $p=0.30$), but the xattn+PE encoder gives -1.57% (hierarchical CI $[-2.4, -0.8]\%$, $p < 10^{-4}$, five seeds both arms). Unlike the no-map xattn+PE cell, this one is *stable* ($\sigma 0.004$): the map both supplies context and tames the encoder’s heavy tail. *Third*, the map-condition benefit is **not** heading re-entry: zeroing the heading-coupled root channel leaves it intact (0.4800 ± 0.0046 , $+0.2\%$ of full pose, $p=0.58$; still -1.55% vs. no-pose+map, paired $p < 10^{-4}$, hierarchical $p=0.07$ at three seeds), and an agent-centric pose frame changes nothing (0.4801 ± 0.0056 , $p=0.65$), ruling out a frame confound. With the map supplying routing and heading, the residual pose signal is *body configuration*. Map context also stabilizes training: per-seed spread collapses ($\sigma \leq 0.006$ vs. up to 0.135; Table 7), and the failure mode of Section 4.2 never occurs in a map-enabled seed.

4.6 ABLATION: BACKBONE PRETRAINING ON FULL WAYMO

The 579-scene subset used above is $\approx 0.12\%$ of the full Waymo training set ($\approx 487\text{k}$ scenarios; 486,995 exactly). To test whether dataset scale is the binding constraint on absolute accuracy, and to

Table 5: Pretrain $\times$ pose ablation, seed-replicated (five seeds per cell). All runs include HD map and use 30-epoch fine-tuning on the 579-scene subset (the no-pretrain rows repeat Table 4’s map cells). Pretrained variants load the same full-Waymo backbone checkpoint via residual zero-init pose fusion; fine-tuning seeds vary.

Pretrain	Pose Enc.	minADE (mean $\pm$ std)	minFDE (mean $\pm$ std)	per-seed minADE
None	—	0.4854 $\pm$ 0.0047	1.0493 $\pm$ 0.0150	.490 .484 .489 .478 .486
None	GRU	0.4823 $\pm$ 0.0058	1.0349 $\pm$ 0.0058	.480 .491 .485 .477 .479
Full Waymo	—	0.3745 $\pm$ 0.0071	0.7915 $\pm$ 0.0159	.382 .375 .380 .371 .364
Full Waymo	GRU	0.3715 $\pm$ 0.0042	0.7863 $\pm$ 0.0137	.374 .376 .365 .371 .371
Full Waymo	XAttn+PE	0.3718 $\pm$ 0.0034	0.7890 $\pm$ 0.0111	.371 .371 .378 .369 .370

isolate pose’s contribution when the backbone already encodes a strong trajectory prior, we run a final 2×2 ablation: $\pm$ pretrain $\times$ $\pm$ pose, with HD map enabled in all four cells.

We pretrain the trajectory-only backbone (no pose encoder) on the full Waymo pedestrian set, 231,507 per-agent training examples drawn from 486,995 scenarios, for 30 epochs with the same hyperparameters as before. To avoid corrupting this backbone at fine-tune iter 0, we use a *residual zero-init* pose fuser: its output is *added* to the backbone feature and its final linear layer is zero-initialized, so the pose pathway contributes exactly zero initially (the backbone passes through unchanged) and warms up additively.

Pretraining is the second dominant lever and is robust: it reduces minADE by 22.9% without pose and 23.0% with pose ($p < 10^{-4}$), by far the largest effect after map, and, like map, trajectory-only. The pose contribution *on top*, however, does not survive five seeds. At three seeds it looked like -1.9% ; at five seeds in both arms the gap shrinks to -0.7% – -0.8% for both encoders, with hierarchical bootstrap $p=0.35$ (xattn+PE) and $p=0.45$ (GRU), neither significant. The two encoders are themselves indistinguishable once pretrained ($+0.1\%$, $p=0.97$), so the from-scratch encoder distinction vanishes against a strong prior. The residual zero-init fuser keeps every pretrained cell stable (no failure seeds), but pretraining, like the no-map condition, does not establish a pose benefit; only the map $\times$ xattn+PE cell does. Qualitative examples are in Appendix A.7, Figure 4.

5 ANALYSIS AND INSIGHTS

How an inert loss can look like a headline result. The failure chain is worth spelling out because none of its links is exotic: a data condition silently degenerates (all-zero future-pose targets), the loss evaluates to a smooth constant with *zero* gradient rather than crashing, two functionally identical configurations get run once each, and the resulting 0.024 seed gap reads as a 3.8% design effect that invites a mechanistic story. Our audit generalizes to two cheap checks that catch it before publication: verify the *target coverage* of every auxiliary loss against the data, and check the *gradient* of every loss term in isolation on a real batch.

Where and how the pose branch helps. Where pose helps (the map $\times$ attention cell), the signal comes from *encoding* past body configuration, not from supervision or heading: it survives zeroing the heading-coupled root channel and an agent-centric frame change (Section 4.5). Real dense future-pose targets, the only condition in which the rotation-space losses carry gradient, do not move accuracy (Appendix A.5), and no auxiliary-loss composition separates from the no-auxiliary control. In the no-map regime, by contrast, nothing is resolvable: heavy-tailed failures (9 of 35 runs, 26%, reach minADE ≥ 0.70 ; bad basins, not late divergence) strike GRU and attention cells alike and vanish with map context or pretraining, which reinforces that on small benchmarks per-seed values or failure rates must be reported, not seed means alone.

6 LIMITATIONS

Dataset scale, and why we cannot go bigger. Our ablations use a 579-scene subset, and SMPL pose pseudo-ground-truth exists *only* for this subset. The full-Waymo backbone we pretrain is necessarily

trajectory-only, so pose-conditioning at full scale is infeasible with current annotations, and whether denser or cleaner pose labels at scale would change the picture is untestable here.

Pose provenance and decoder. The LiDAR-derived SMPL pseudo-GT (Waymo-3DSkelMo) couples the root-orientation channel to past trajectory (Section 4.1). The surviving map-condition benefit is robust to zeroing that channel (Section 4.5), but the body-joint estimates themselves are pipeline outputs, not ground truth; deployed online pose estimation would carry different noise, and we do not evaluate predicted-pose quality (our metric is trajectory only).

Statistical power and two bootstrap levels. Our primary paired bootstrap resamples pedestrians on *seed-averaged* errors; we also report a *hierarchical* bootstrap that resamples seeds then pedestrians, which is the conservative test we rely on. Under it the map (-26%) and pretraining (-22%) trajectory effects survive strongly ($p < 10^{-4}$), and exactly one pose effect does, map $\times$ xattn+PE at -1.6% ($p < 10^{-4}$), while every other pose margin does not (no-map xattn+PE $p=0.83$, GRU+map $p=0.30$, pretrained pose $p \geq 0.35$); this is precisely the discipline that separates the one genuine pose effect from the several that a three-seed or single-run evaluation would have reported as real (Sections 4.2–4.3). We test ~ 30 pairs uncorrected: the relied-upon effects ($p \leq 10^{-4}$) survive Bonferroni at this family size; the 0.01–0.05 range is exploratory. The best-checkpoint protocol selects on validation noise; all CIs cover one validation split only.

7 CONCLUSION

We have presented MTR+POSE, a pose-conditioned extension of the Motion Transformer over past SMPL body poses (6D rotations), together with a seed-replicated measurement of what pose conditioning is worth. Our conclusions: (1) Pose helps in a well-defined setting: the cross-attention+PE encoder *with* HD map improves minADE by 1.6%, stable across five seeds ($p < 10^{-4}$). Control ablations (root-orientation zeroing, agent-centric frame) show this benefit is genuine body-configuration signal, not heading re-entry, and the mechanism is past-pose *encoding* rather than future-pose prediction. (2) This benefit is specific to the map condition and the attention encoder. The GRU does not separate with map, and no pretrained pose benefit survives five seeds. In the no-map condition *no* configuration reliably beats the no-pose baseline: cross-attention+PE looks like a stable -3.5% at three seeds, but both further seeds fail (0.6648 ± 0.0381 , $+0.7\%$, n.s.), and the apparent positional-encoding ordering effect dissolves with it. (3) Establishing this boundary required seed replication: at three or fewer seeds, several of the effects above would have been reported as real. The same discipline led us to an auxiliary-loss finding of independent interest: a widely used geodesic SO(3) supervision loss is inert under sparse future-pose annotation, receiving zero gradient against absent future-pose targets, so apparent effects attributed to it in single-run comparisons (up to 7.6%) are seed noise between what are, in fact, identical configurations. The replication and auxiliary-loss-auditing protocol, alongside the pose result itself, is our intended lasting contribution.

REPRODUCIBILITY STATEMENT

All experiments use the publicly available Waymo Open Motion Dataset and the open-source Waymo-3DSkelMo pose-extraction pipeline; neither is redistributed by us. Training configurations (YAML), the multi-seed launch script, the per-pedestrian metric logger, and the bootstrap analysis script are included in our code release, together with the exact random seeds (101/202/303, plus 404/505 for the headline, encoder, map, and pretrain cells) and the per-epoch validation protocol described in Section 4.1. Appendix A lists all hyperparameters; Appendix A.2 documents dataset construction, including the timestamp-alignment and future-pose-coverage details on which the inert-gradient analysis rests. We will release the code upon publication.

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LLM DISCLOSURE

The authors used large language model (LLM) assistance (Claude, Anthropic) during the preparation of this manuscript for writing drafts, code assistance, and literature search support. All experimental results, scientific analysis, and conclusions are the authors' own.

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A ADDITIONAL EXPERIMENTAL DETAILS

A.1 ARCHITECTURE HYPERPARAMETERS

- GRU pose encoder: GRU(input=144, hidden=256, layers=2, dropout=0.1); $\approx 505\text{K}$ parameters total including fusion MLP
- Cross-attention pose encoder: pose projection 144 $\rightarrow$ 256, MultiheadAttention(embed_dim=256, num_heads=8, dropout=0.1, batch_first=True); $\approx 497\text{K}$ parameters total including fusion MLP
- Pose input dimension: 144 (24 joints $\times$ 6D rotation)

- Past pose window: 11 timesteps (1.0 second at 10 Hz)
- Number of predicted trajectory modes: $K = 6$
- Number of future steps: 80 (8.0 seconds)
- MTR context encoder: 6 local-attention transformer layers, 256 hidden dim, 8 heads, attention dropout 0.1
- MTR decoder: 6 layers, hidden dim 512 (map branch 256), 8 heads
- Batch size: 10
- Optimizer: AdamW, $\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 0.01
- LR schedule: initial LR 1×10^{-4} , step decay $\times 0.5$ at epochs 10, 20, 25
- Epochs: 30
- Gradient clipping: $\text{norm} \leq 1.0$
- Random seeds for replicated runs: 101, 202, 303 (plus 404, 505 for headline cells)

A.2 DATASET CONSTRUCTION

Waymo agent future trajectories are stored in `processed_scenarios/` PKL files with shape `(num_agents, 91, 10)`. SMPL pedestrian files are matched to Waymo object IDs via integer stem matching. 6,688 of 8,282 validation pedestrians (80.8%) have a valid future-trajectory match and are evaluated; the remainder are excluded. The 91-step grid spans $[0, 9.0]$ seconds; SMPL timestamps (originally in microseconds) are divided by 10^6 before alignment. In the 10 fps condition the SMPL observations cover only $[0, 1.0]$ s (the past window): across the entire training split only 5 of 91,520 future pose-target steps are nonzero, which is the basis for the inert-geodesic-gradient analysis in Section 4.3.

A.3 LOSS WEIGHT STABILITY

At $w = 1.0$, gradient explosion through the SMPL forward kinematics causes NaN divergence at epoch 3. The geodesic loss gradient norm was also found to diverge at the ± 1 boundary of the $\arccos$ argument; we clamp to $[-1 + \epsilon, 1 - \epsilon]$ with $\epsilon = 10^{-7}$. At $w = 0.1$, training is stable for all 30 epochs across all ablation variants.

A.4 AUXILIARY LOSS WEIGHT

A single-seed sweep, nominally over the geodesic weight, appears to find a “sharp optimum” at $w=0.1$: $w=0.05$ gives 0.6786 and $w=0.2$ gives 0.6660. Two caveats apply. First, because the geodesic term is inert, the sweep actually varies the *WTA-L1* weight (both weights were scaled together in those configurations). Second, the differences involved (0.02–0.06) overlap the seed-noise scale measured in Section 4.3, so the curvature of this one-seed sweep should not be read as a calibrated sensitivity estimate. What survives: $w=0.1$ is a reasonable default, and we did not find evidence that performance is improved by halving or doubling it.

A.5 ABLATION: FUTURE POSE SUPERVISION QUALITY

To localize the source of improvement, we compare two data conditions, both with the full auxiliary loss set at $w=0.1$, **seed-replicated (five seeds each)**: (a) *no future poses*, where future-pose GT is zero and only past poses are observed (the inert-loss condition of our other ablations): minADE 0.6618 ± 0.0366 ; and (b) *30 fps motion-prior-completed future poses*, where the Waymo-3DSkelMo pipeline infills 99% of future timesteps as pseudo-GT, the one setting in which the geodesic and MPJPE terms receive real, non-degenerate gradients: minADE 0.6586 ± 0.0292 .

Activating the future-pose losses changes minADE by -0.5% (hierarchical bootstrap $p=0.60$; vs. the no-pose baseline -0.3% , $p=0.82$): **not significant**. Giving the rotation-space losses genuine gradients does not reliably improve trajectory accuracy over the inert condition. This is consistent with our single-run pilot (0.6567 vs. 0.6532), and both conditions carry the same heavy-tailed seeds as every other no-map cell. Whatever benefit pose conditioning provides comes from the past-pose encoder, not the future-pose prediction task.

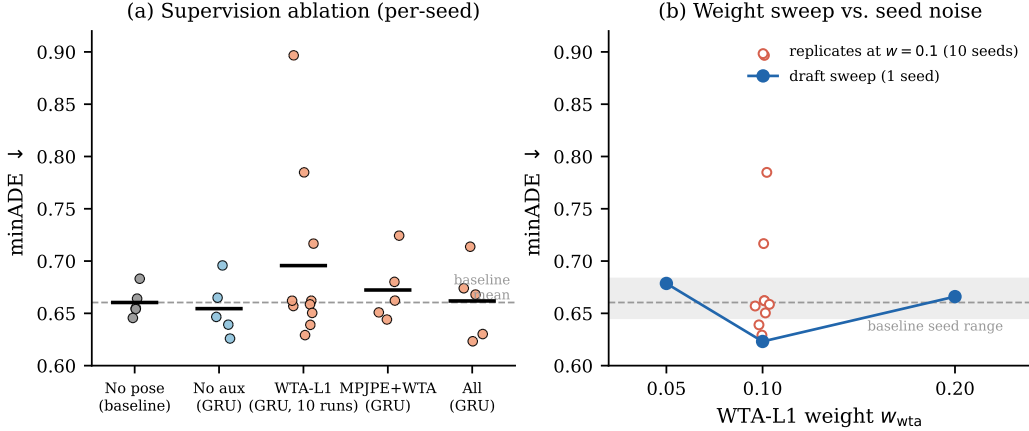


Figure 2: **Loss ablation and WTA-L1 weight sensitivity.** *Left:* minADE for the seed-replicated supervision configurations (Table 2); markers show individual seeds. *Right:* single-seed sensitivity to the auxiliary WTA-L1 weight ($w \in \{0.05, 0.1, 0.2\}$), with the seed-noise scale measured in Section 4.3 shown as a shaded band. Within that noise, the apparent optimum at $w=0.1$ is weak evidence at best.

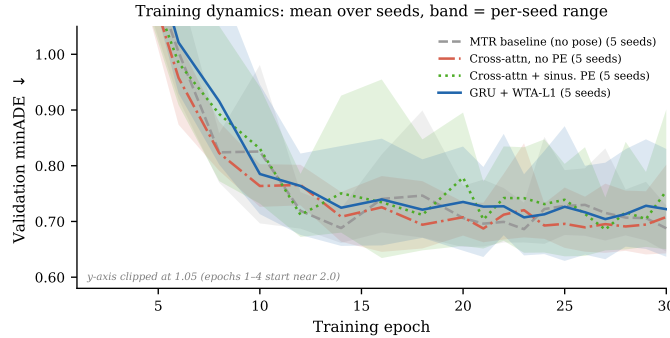


Figure 3: **Training dynamics.** Validation minADE over training for the temporal encoding variants; lines are means over seeds, bands the per-seed range. Epoch-to-epoch and seed-to-seed variability is of the same order as the differences between several variants, which is why all comparisons in this paper are seed-replicated.

A.6 TRAINING DYNAMICS

Figure 3 shows validation minADE over 30 training epochs for the temporal encoding variants, with per-seed bands (five seeds each). Two qualitative observations are stable across seeds: cross-attention without positional encoding tracks the trajectory-only baseline throughout, and the pose-conditioned variants exhibit larger epoch-to-epoch variability than the baseline. The failure mode is a bad basin, not a late-training collapse: both the GRU’s failed seeds (plateaus near 0.72–0.83) and the cross-attention+PE failure seed (505) sit high from early in training. At five seeds the xattn+PE band is no longer the narrowest; its three-seed tightness was the sampling artifact discussed in Section 4.2.

A.7 QUALITATIVE EXAMPLES

A.8 ORIGINAL SINGLE-RUN SWEEP

Table 6 reports all 19 single-seed exploratory runs that motivated this study’s ablation design, with loss labels corrected per the audit: “geo” terms were inert (zero gradient, Section 4.3), so the functional supervision is what remains after removing them. Runs 009 and 010 are functionally



Figure 4: **Qualitative comparison at full-Waymo-pretrained scale.** Four validation scenes. Past in gray; GT future dashed black with $\star$ endpoint. All 6 predicted modes faded; top-scoring mode highlighted (red: no-pose; blue: pose). Per-scene minADE in titles. Scenes selected from the largest per-pedestrian improvements of the pose model, so these illustrate the failure modes pose can fix, not typical magnitudes. *Long-distance*: no-pose stops at 5m; pose follows the 25m walk. *Curved*: no-pose underestimates displacement; pose follows the curve. *Mode coverage*: both top modes underestimate, but pose has a faded mode aligned with GT. *Stationary*: no-pose wrongly predicts motion; pose correctly stays near origin.

identical; their spread is seed noise. These runs use unrecorded seeds and are superseded by the seed-replicated matrix of Table 7; we report them for completeness and to show the single-run values that a non-replicated ablation would have reported.

Table 6: Original single-run sweep (corrected labels). “Functional losses” removes inert terms; w in parentheses. Baseline for Δ is run 004. Runs 001–002 are invalid (zero future trajectory GT).

Run	Label	Encoder	Functional losses / Map / Pretrain	minADE	Δ
001	all losses, $w=1.0$	GRU	all (1.0)	0.6114	<i>invalid</i>
002	no pose	—	traj only	0.7724	<i>invalid</i>
003	MPJPE+WTA-L1+cls, $w=0.1$	GRU	mpjpe+wta+cls (0.1)	0.6532	−3.2%
004	baseline	—	traj only	0.6745	—
005	MPJPE+WTA-L1+cls, $w=0.05$	GRU	mpjpe+wta+cls (0.05)	0.6557	−2.8%
006	MPJPE+WTA-L1+cls, $w=0.2$	GRU	mpjpe+wta+cls (0.2)	0.6542	−3.0%
007	full loss set, 30fps future	GRU	mpjpe+geo+wta+cls (0.1), real future	0.6567	−2.6%
008	MPJPE+WTA-L1, $w=0.1$	GRU	mpjpe+wta (0.1)	0.6576	−2.5%
009	WTA-L1, $w=0.1$ (seed A)	GRU	wta (0.1)	0.6467	−4.1%
010	WTA-L1, $w=0.1$ (seed B)	GRU	wta (0.1) [geo inert]	0.6231	−7.6%
011	WTA-L1, $w=0.05$	GRU	wta (0.05) [geo inert]	0.6786	+0.6%
012	WTA-L1, $w=0.2$	GRU	wta (0.2) [geo inert]	0.6660	−1.3%
013	cross-attn, no PE	XAttn	wta (0.1) [geo inert]	0.6765	+0.3%
014	cross-attn + PE	XAttn+PE	wta (0.1) [geo inert]	0.6337	−6.0%
015	no pose, with map	—	traj only + map	0.4880	−27.7%
016	pose, with map (GRU)	GRU	wta (0.1) + map [geo inert]	0.4797	−28.9%
017	pretrain backbone	—	traj only + map (full Waymo, 231k peds)	—	<i>ckpt only</i>
018	finetune, pose (GRU)	GRU	wta (0.1) + map, pretrained	0.3749	−44.4%
019	finetune, no pose	—	traj + map, pretrained	0.3861	−42.8%

A.9 SEED-REPLICATED MATRIX

Table 7 lists every run of the replication matrix (seeds 101/202/303, plus 404/505 for the headline, encoder, map, and pretrain cells). The four audit cells were additionally re-run with the future-pose mask fix (Section 4.3); those masked runs are in the code release.

Table 7: Seed-replicated matrix: best-checkpoint minADE per seed (seeds 101/202/303, plus 404/505 for headline cells). All runs 30 epochs, identical protocol; “geo (inert)” marks the zero-gradient geodesic term carried by some configurations.

Cell	Configuration	minADE mean $\pm$ std	per-seed values
baseline	no pose	0.6604 ± 0.0143	.6550 .6831 .6641 .6541 .6456
geo_pure	GRU, no active aux (geo inert)	0.6545 ± 0.0271	.6958 .6650 .6392 .6259 .6466
wta01	GRU, WTA-L1 + geo (inert)	0.6853 ± 0.0651	.6390 .7848 .6292 .7167 .6570
gmm_only	GRU, WTA-L1 (same config as wta01)	0.7060 ± 0.1067	.8967 .6621 .6621 .6586 .6504
mpjpe	GRU, MPJPE + WTA-L1	0.6723 ± 0.0321	.7243 .6622 .6441 .6801 .6509
full	GRU, MPJPE+WTA+cls (geo inert)	0.6618 ± 0.0366	.6680 .6740 .6302 .6233 .7137
xattn	XAttn no PE, WTA-L1 (+ geo inert)	0.6705 ± 0.0336	.7136 .6408 .6457 .6526 .6996
xattn_pe	XAttn+PE, WTA-L1 (geo inert)	0.6648 ± 0.0381	.6395 .6343 .6374 .7098 .7029
norootorient	XAttn+PE, root-orient zeroed	0.6628 ± 0.0056	.6627 .6666 .6605 .6548 .6694
pose30fps	GRU, full loss, 30 fps future poses	0.6586 ± 0.0292	.6559 .6542 .6338 .6408 .7081
map_nopose	no pose, with map	0.4854 ± 0.0047	.4903 .4839 .4886 .4783 .4860
map_wta01	GRU, WTA-L1, with map	0.4823 ± 0.0058	.4798 .4911 .4853 .4768 .4787
map_xattn_pe	XAttn+PE, WTA-L1, with map	0.4778 ± 0.0041	.4769 .4811 .4795 .4710 .4804
map_norootorient	XAttn+PE, root zeroed, map	0.4800 ± 0.0046	.4766 .4853 .4782
map_agentrot	XAttn+PE, agent-centric pose, map	0.4801 ± 0.0056	.4753 .4862 .4787
ft_nopose	no pose, map, pretrained	0.3745 ± 0.0071	.3815 .3753 .3804 .3706 .3645
ft_wta01	GRU, WTA-L1, map, pretrained	0.3715 ± 0.0042	.3744 .3760 .3651 .3711 .3709
ft_xattn_pe	XAttn+PE, WTA-L1, map, pretrained	0.3718 ± 0.0034	.3710 .3712 .3777 .3694 .3697